

Well-Wall-Wave Theory: Orbital Structure from a Composite Field

TZPID Companion Spine Series, Paper C7 of XXVII
Gaussian wells, Bessel waves, and derived stable shells

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Abstract

This companion develops the Well-Wall-Wave (WWW) theory: a composite field model in which orbital structure arises from a smooth inward potential basin (Well), an oscillatory radial eigenmode structure (Wave), and a derived set of dynamically stable shells (Wall). The source manuscript’s stated thesis: “We propose the Well–Wall–Wave (WWW) Theory, a composite field model in which orbital structure arises from the interaction of (i) a smooth inward potential basin (Well), (ii) an oscillatory radial eigenmode structure (Wave), and (iii) a derived set of dynamically stable shells (Wall). The Well is modeled by a Gaussian deficit envelope, while the Wave is modeled via Bessel functions representing radial resonances.” The paper organizes this thesis as a TZPID gold spine: a curated seven-node registry chain (ID9516, ID7732, ID1863, ID0252, ID6489, ID0261, ID0148) with typed proof-obligation roles, dictionary and encyclopedia anchors for every node, a declared dependency order, and stubs in the program’s Isabelle/Wolfram verification pipeline. Within the arc this paper is the mechanism’s planetary-scale showcase: the enclosure’s cavity physics generating orbital architecture.

1 Introduction

The Well-Wall-Wave theory is the most pictorial member of the corpus and among the most consequential: it proposes that orbital structure—the spacing of stable shells around a central mass—is cavity physics. Three components compose the model. The *Well* is a smooth inward potential basin, modeled by a Gaussian deficit envelope (the deficit language of companion C1 made gravitational). The *Wave* is an oscillatory radial eigenmode structure, modeled by Bessel functions representing radial resonances—the same j_ℓ ladder that organizes every cavity in the core series. The *Wall* is not imposed: it is the derived set of dynamically stable shells where well and wave conspire, the radii at which orbiting matter accumulates.

The registry already carries the WWW machinery in its quantization entries: the Well/Wall/Wave cavity selector (ID0252) and its harmonic ladder (ID0201) were load-bearing nodes of core Papers II and IV. This companion gives those nodes their source theory and their astronomical ambition: if shell radii are Bessel-ladder radii dressed by a Gaussian well, then orbital architectures are spectra, and the discrete rung-selection observed from bedforms to resonances (Papers I–II) extends to the layout of planetary systems themselves.

The spine below assembles the composite model’s registry chain. Its sharpest refinement target is a confrontation: fit the derived shell formula against measured exoplanetary system architectures, with the well depth and wave number as the only parameters.

2 The concept-anchored spine methodology

The companion series applies a uniform method, stated here so that the paper is self-contained. The source manuscript was published with its mathematics rendered as text rather than as machine-readable LaTeX, so its spine is *concept-anchored*: the thesis is taken from the manuscript’s abstract, and the spine nodes are the canonical-registry entries most strongly matched to the manuscript’s themes, selected by weighted term overlap with foundational nodes ranked first. Each node carries its registry equation, its dictionary definition, its encyclopedia interpretation, and a declared proof-obligation role (Core_Definition, Core_Axiom, or Derived_Theorem_Obligation). The resulting chain is a starting backbone for refinement—a typed scaffold onto which an exact, equation-by-equation crosswalk with the source manuscript can be progressively attached—rather than a claim of completed formalization.

Three properties make the backbone useful even before refinement completes. First, *addressability*: every claim in the companion paper resolves to a registry identifier whose equation, definition, and verification status are independently inspectable, so disagreement can be localized to a node rather than to the paper as a whole. Second, *pipeline coupling*: each spine is paired with an Isabelle focus-theory stub and a Wolfram check stub, so that as nodes are refined their obligations enter the same machine-checked pipeline that certifies the core series’ guardrails. Third, *role discipline*: the obligation roles separate what the spine defines from what it asserts and what it must prove, which keeps the demarcation section of every companion paper honest by construction.

The dependency chain should accordingly be read as a proof-order proposal: upstream nodes supply the vocabulary and axioms that downstream nodes consume, and the refinement pass is free to reorder where the source manuscript’s own logic demands it. Where a node’s registry equation arrives with rendering artifacts inherited from the source extraction (a known cost of text-rendered mathematics), the equation is quoted as carried by the registry, with the dictionary and encyclopedia entries supplying the authoritative reading.

3 The registry chain

The curated chain, in proof order, is

$$\begin{aligned} \text{ID9516} &\rightarrow \text{ID7732} \rightarrow \text{ID1863} \rightarrow \text{ID0252} \\ &\rightarrow \text{ID6489} \rightarrow \text{ID0261} \rightarrow \text{ID0148}. \end{aligned}$$

We take the nodes in order; each subsection quotes the registry equation as carried by the canonical master, then gives the dictionary definition and the encyclopedia’s interpretive reading.

3.1 ID9516: Gravitation: Spacetime Curvature and Lensing Anomalies: Phi_eff (Core_Definition)

Registry equation:

$$\backslash\text{Phi}_{\{\text{eff}\}} = (1+\epsilon)\backslash\text{Phi}_{\{\text{Newton}\}}$$

Dictionary definition. Gravitational lensing provides one testing ground.

Encyclopedia reading. Gravitation: Spacetime Curvature and Lensing Anomalies: Phi_eff is an algebraic definition, written $\text{Phi_eff} = (1+\epsilon)\text{Phi_Newton}$. It is an algebraic definition expressing the quantity directly in terms of the variables on its right-hand side. It is built from epsilon, defining Phi_eff. Within the Trawin Zero Point registry it serves as one element of the framework’s mathematical narrative.

Computational guardrail: well_wall_wave_theor_ID9516_identity.

Obligation reading. As a Core_Definition this node contributes vocabulary rather than assertion: downstream nodes may use its object freely, but nothing is yet claimed about the world. Its refinement burden is precision—the typed form must capture exactly the object the source manuscript deploys.

3.2 ID7732: How This Ties into Your TZPQVS Superconducting Spherical Enclosure + Cymatics Id (1) (Core_Definition)

Registry equation:

$$Y_{\{\ell\ m\}}(\theta, \phi)$$

Dictionary definition. Relation: $Y_{\ell m}(\theta, \phi)$

Encyclopedia reading. How This Ties into Your TZPQVS Superconducting Spherical Enclosure + Cymatics Id (1) is a symbolic expression, written $Y_{\ell m}(\theta, \phi)$. It introduces a symbol or relation used elsewhere in the framework. It is built from ℓ , θ , ϕ , Y_m . Within the Trawin Zero Point registry it serves as one element of the framework's mathematical narrative.

Computational guardrail: `well_wall_wave_theor_ID7732_structure`.

Obligation reading. This Core_Definition fixes an object the rest of the chain consumes. In proof order it must be discharged first; in refinement it is where a mismatch between the manuscript's usage and the registry's typing would first surface.

3.3 ID1863: The Inward-Outward Cross (Core_Definition)

Registry equation:

$$P(r) = P_{\infty} - \frac{1}{2} \rho v(r)^2$$

Dictionary definition. The Inward-Outward Cross — $P(r) = P_{\infty} - (1/2)\rho v(r)^2$ [ID1863]

Encyclopedia reading. gical experience of cycling through the Trawin Hamiltonian's components.** The "breakthrough" sensation at the orthogonal collision is the experience of $\hat{H}_{\text{TZP}}$ engaging — the zero-point extraction, dimensional collapse, and infinite-order transition activating at the crossing point. — ## The Pressure Sink and the TZP Potential (ID 023) The pressure sink at the centre of the cross drain — the low-pressure region that draws fluid inward — is the fluid-dynamic realisation of the TZP potential (ID 023: $V_T(x) = \lambda e^{-d(x,p)/u_T}$). The pressure field around the drain: decreases toward the centre as the velocity increases. Near the central hole, the pressure drops sharply — analogous to the exponential deepening of the TZP potential well near p_0 . **The radial acceleration $a_r = -(1/\rho)\partial P/\partial r$ is the fluid analogue of the gravitational acceleration g

Computational guardrail: `well_wall_wave_theor_ID1863_identity`.

Obligation reading. A definitional node: its function is to make later statements well-formed. The guardrail attached to it is correspondingly structural—an identity or well-formedness check rather than a physical claim.

3.4 ID0252: Harmonic Bessel Standing Waves (Core_Definition)

Registry equation:

$$u_{\{mn\}}(r, \theta) = J_m(k_{\{mn\}}r) e^{im\theta} \quad || \quad J_m(k_{\{mn\}}R) = 0$$

Dictionary definition. Resonant frequencies satisfy $f_n = n f_{n-1}$ for integer n .~In three-dimensional cavities the standing wave patterns are given by Bessel functions $J_n(kr)$; nodal radii occur at the zeros of J_n and organise matter and energy into stable patterns.

Encyclopedia reading. The harmonic series and Bessel functions explain why only certain frequencies and radial modes appear in a resonant TZP cavity. Interpretive role: This entry specifies the structure within the registry.

Computational guardrail: well_wall_wave_theor_ID0252_identity.

Obligation reading. As a Core_Definition this node contributes vocabulary rather than assertion: downstream nodes may use its object freely, but nothing is yet claimed about the world. Its refinement burden is precision—the typed form must capture exactly the object the source manuscript deploys.

3.5 ID6489: \{Einstein–Rosen Bridge and Trawin Zero-Point Tunnel:\} (Derived_Theorem_Obligation)

Registry equation:

0.3em] \textit{A Theoretical Massless Transit System of Attainable Thermodynamics}\Through Wave and Inter-State Topology} } \author{ Daniel Alexander Trawin\small ORCID: \href{https://715}{0009-0001-4630-3715}\small \texttt{daniel.alexander.trawin@placeh ...

Dictionary definition. \{Einstein–Rosen Bridge and Trawin Zero-Point Tunnel:\} — 0.3em] textitA Theoretical Massless Transit System of Attainable ThermodynamicsThrough Wave and Inter-State Topology author Daniel Alexander Trawin[0.

Encyclopedia reading. \{Einstein–Rosen Bridge and Trawin Zero-Point Tunnel:\} is a tbd, written 0.3em] textitA Theoretical Massless Transit System of Attainable ThermodynamicsT. Its construction method could not be determined from the available material. Within the Trawin Zero Point registry it serves as one element of the framework’s mathematical narrative.

Computational guardrail: well_wall_wave_theor_ID6489_identity.

Obligation reading. A derived obligation: the spine claims this follows from what precedes it. Until the formal derivation is supplied, the computational guardrail polices at least the node’s internal consistency.

3.6 ID0261: Bessel Standing Wave Quantization (Derived_Theorem_Obligation)

Registry equation:

approx1.2024 || frac{R}{r}\approx2.46

Dictionary definition. In a cavity with cylindrical or spherical symmetry, allowed radial modes satisfy Bessel’s equation; quantized modes correspond to zeros of $J_n(kr)$, dictating allowable radii and frequencies for standing waves803815022944855L32-L43.

Encyclopedia reading. The radial structure of resonant modes is not arbitrary; it is fixed by the zeros of Bessel functions, leading to discrete energy levels. Interpretive role: This entry specifies the structure within the registry.

Computational guardrail: well_wall_wave_theor_ID0261_structure.

Obligation reading. This node carries proof debt by declaration—Derived_Theorem_Obligation—and is therefore where the refinement pass should concentrate first: a discharged derivation here strengthens every downstream node simultaneously.

3.7 ID0148: Phase Twist-Induced Domain Wall (Derived_Theorem_Obligation)

Registry equation:

$\phi(x)=\pi\left[\frac{1+\tanh((x-x_0)/w)}{2}\right] || \Delta\phi = \phi(+\infty)-\phi(-\infty)=\pi$
 $|| \nu_{\mathrm{wall}}=\frac{1}{2\pi}\int \partial_x\phi, dx$

Dictionary definition. Introducing a smooth phase ramp over the Gaussian window creates a domain wall connecting two frequency states.

Encyclopedia reading. Technical interpretation remains tied to the local registry and equation witness. Interpretive role: This entry gives the bridge operations a topological defect interpretation. It turns the phase-twist machinery into a seam that can persist as a local structural object.

Computational guardrail: `well_wall_wave_theor_ID0148_threshold`.

Obligation reading. As a `Derived_Theorem_Obligation` this node owes a proof: the registry asserts that it follows from the chain’s upstream vocabulary and axioms, and the Isabelle focus stub holds the slot where that derivation will be discharged.

4 Dependency structure and obligations

Table 1 summarizes the spine’s obligation ledger. The chain’s proof order runs from the definitional nodes (vocabulary first) through the derived obligations to the spine’s closure; the refinement pass may interleave additional registry nodes where the source manuscript’s own derivations require them, in which case the table grows monotonically—existing rows are never weakened, only joined.

Table 1: Obligation ledger for the well wall wave theory spine.

ID	Obligation role	Wolfram check
ID9516	Core_Definition	<code>well_wall_wave_theor_ID9516_identity</code>
ID7732	Core_Definition	<code>well_wall_wave_theor_ID7732_structure</code>
ID1863	Core_Definition	<code>well_wall_wave_theor_ID1863_identity</code>
ID0252	Core_Definition	<code>well_wall_wave_theor_ID0252_identity</code>
ID6489	Derived_Theorem_Obligation	<code>well_wall_wave_theor_ID6489_identity</code>
ID0261	Derived_Theorem_Obligation	<code>well_wall_wave_theor_ID0261_structure</code>
ID0148	Derived_Theorem_Obligation	<code>well_wall_wave_theor_ID0148_threshold</code>

5 Crosswalk to the core series and companion corpus

Because all spines draw on one canonical registry, node sharing is measurable and meaningful: a shared identifier means two papers consume literally the same typed object, so verification work transfers between them without translation. Of this spine’s seven nodes, 3 appear in core-series spines and 1 recur elsewhere in the companion corpus. Table 2 records the census.

Two readings follow. Nodes shared with the core series are this spine’s strongest members: their guardrails are already certified in the core pipeline, and the present paper inherits that status. Nodes unique to this spine are its distinctive contribution—and its refinement priority, since no other paper will discharge them. Nodes recurring across companions mark the corpus’s internal seams: the same object consumed by several manuscripts, whose refinements must agree by the duplicate discipline documented in the Einstein Focus companion.

6 Position in the TZPID arc

WWW is the source theory behind ID0252 and ID0201, which the core series consumed as given: Paper II’s quantization link and Paper IV’s acoustic drive both trace here. Upward, the model is Paper I’s enclosure acoustics turned gravitational—the filament-web argument at stellar-system scale; downward, its stable shells are the capture targets of Paper II’s tidal locking.

Table 2: Node-sharing census for this spine.

ID	Core-series appearances	Companion-corpus appearances
ID9516	—	—
ID7732	I (Nested Hyperspherical Enclosures), II (Phase Locking)	—
ID1863	—	—
ID0252	II (Phase Locking), IV (Gyro-magnetic Movement)	—
ID6489	—	Gravitational-Emergence-TPZ-; theory of it all trawin topology; trawin enlil protocol (+2 more)
ID0261	II (Phase Locking)	—
ID0148	—	—

7 Verification pathway

The spine enters the program’s two-engine verification pipeline. The Isabelle focus theory `TZPID_Spine_well_wall_wave_` types the seven obligations over the registry’s declared vocabulary, in the dependency order of Section 3; the Wolfram check stub `well_wall_wave_theory_checks.wl` carries the per-node computational guardrails listed in Table 1. As with the core series, the division of labor is deliberate: the theorem prover certifies structure (types, roles, dependency soundness), while the computational engine certifies arithmetic and algebraic identities node by node. A check name of the form `..._identity` denotes a per-node identity guardrail generated from the registry equation; refined checks replace these as the equation-level crosswalk matures.

Because the companion spines share the registry with the core series, verification is cumulative: any node here that also appears in a core spine (the chain tables record several) inherits the core guardrail’s status, and any refinement proved here propagates back. The companion series thus functions as the proof pipeline’s breadth dimension—161 distinct registry nodes across the 27 companions—complementing the core series’ depth.

The pipeline’s two-engine division also fixes what failure means at this layer. A failed Wolfram guardrail localizes an arithmetic or algebraic defect in one node’s registered equation: the repair is editorial (correct the registration against the source) or substantive (the source itself errs), and the obligations ledger records which. A failed Isabelle discharge, by contrast, indicts the chain: a type mismatch or unprovable dependency means the proof-order proposal of Section `refsec:chain` is wrong as stated, and the refinement pass must re-curate. Keeping these failure modes separate is what the stub architecture buys: every companion enters the pipeline with its failure semantics declared in advance, which is the practical meaning of being peer-reviewable by machine.

8 Refinement worksheet

The concept-anchored backbone converts into an equation-level formalization through a bounded, node-by-node worksheet. For each node the worksheet item below states the concrete refinement act; completing all seven closes the spine’s first formalization pass.

1. **ID9516** (`well_wall_wave_theor_ID9516_identity`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.

2. **ID7732** (`well_wall_wave_theor_ID7732_structure`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.
3. **ID1863** (`well_wall_wave_theor_ID1863_identity`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.
4. **ID0252** (`well_wall_wave_theor_ID0252_identity`): exhibit the source manuscript’s working definition beside the registry equation; reconcile notation; type the object in the focus theory and confirm every downstream node consumes the typed form.
5. **ID6489** (`well_wall_wave_theor_ID6489_identity`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
6. **ID0261** (`well_wall_wave_theor_ID0261_structure`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.
7. **ID0148** (`well_wall_wave_theor_ID0148_threshold`): reconstruct the source manuscript’s derivation sketch; identify the upstream nodes it actually uses (amending the chain if they differ from the curated order); discharge the proof in the focus theory or record the precise gap.

The worksheet’s order matters: definitional items unblock derived ones, and a failure at any item halts propagation rather than silently weakening downstream claims. Completed items should update the obligations CSV in place, so that the spine’s public ledger always reflects its true refinement state—the same live-ledger discipline the core series applies to its guardrails.

9 Demarcation and refinement targets

As throughout the program, the demarcation line is drawn explicitly. The established layer comprises the standard mathematics and physics that the chain’s nodes quote—definitions, identities, and independently citable results—together with whatever the computational guardrails certify. The framework layer comprises the source manuscript’s interpretive claims and the spine’s structural assertion that its nodes form the stated dependency chain. The composite decomposition is a modeling choice; the falsifiable surplus is the derived shell spectrum, which either matches measured orbital architectures at stated parameter counts or does not.

Refinement targets, in pipeline order: (i) replace the concept-anchored node selection with an equation-level crosswalk against the source manuscript; (ii) promote each `..._identity` guardrail to a content-bearing check; (iii) discharge the typed obligations in the Isabelle focus theory against the program’s shared vocabulary; and (iv) record any node whose refinement fails, since a failed node localizes exactly where the source manuscript and the canonical registry disagree—the information a refinement pass exists to produce.

10 Conclusion

Well-Wall-Wave theory bets the enclosure picture at the scale where humanity has the best data: orbital architecture. Its components are each standard—Gaussian wells, Bessel resonances, stability analysis—and its composite is a spectrum generator with two parameters. The refinement program writes itself: derive the shell formula’s predictions for the catalogued exoplanet systems and count the successes against the parameter cost. Few claims in the corpus are as directly exposed to existing data.

References

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